

Week 6

Aim

The aim of this seminar was to understand the techniques of static malware analysis in a safe way using only windows executables to understand how analysts look at suspicious files before engaging with them in analysis.

Glossary

Static Analysis – Examining a program without executing it.

Portable executable file – Standard file format for windows executables. They contain metadata and inspecting this can reveal how a program is built.

String extraction – Static analysis technique that reads ASCII sequences from a binary file that may reveal sensitive information about the malware.

Entry point – Memory inside a PE file where the programs execution begins.

Yara rule - A detection signature that identifies patterns within a file, often used to classify malware families.

Triage workflow - A sequence of analysis steps to access whether a file is suspicious or not.

Errors and Solutions

Within this lab, I encountered import errors when trying to use PE header analysis and YARA matching and this caused the virtual environment to crash. Therefore, I then created a clean virtual environment for week 6 tasks and reinstalled pefile and yara python from scratch which solved module conflicts.

Another issue I had was that python was throwing UnicodeDecodeError due to invalid byte sequences and therefore I added errors=ignored which skips the unreadable characters to stop the script from crashing.

Reflection

During this lab, compared to other seminars, I felt like this was where most of my weak points were located. I depended a lot on the code and instructions provided to me in the lab worksheet too complete this lab work. One key reflection I have is that if I knew more, I felt like I could have provided a much more interesting challenge rather than a simple http string.

Key Takeaways

Static Analysis helps a lot – A lot of information can be learnt about a file without ever running it, and a lot of metadata provides strong clues about the behaviour of the program

Yara – Yara is a powerful tool to help easily identify patterns that can be detected across binaries and therefore it is critical in more advanced situations where in which malware is more sophisticated and therefore needs more precise classification.